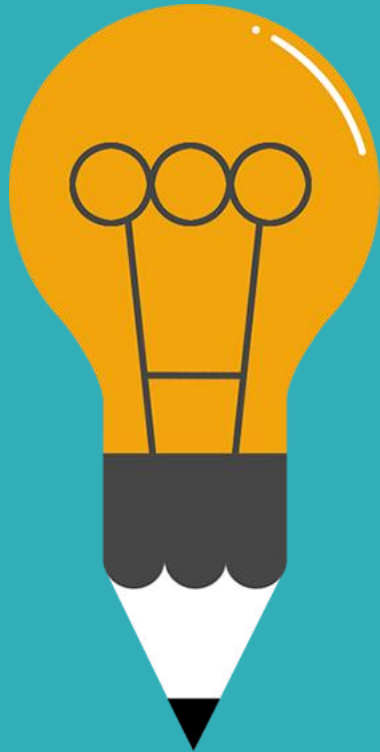
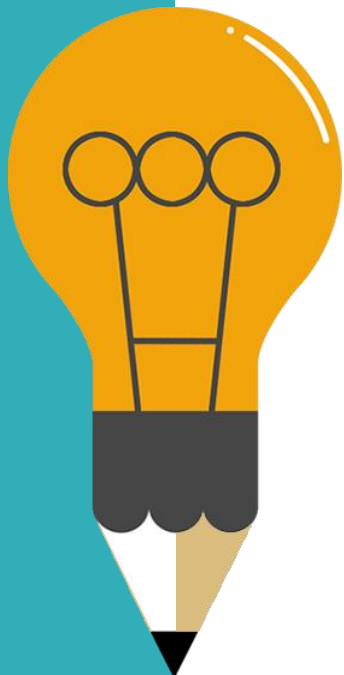




Adult Income Classification Model

Project Outline



01 Data Exploration

02 Data Cleaning

03 Feature Engineering

04 Preprocessing Pipeline

05 Modeling

06 Deployment & Prediction

Real Life Application

Takaful and Karama Program



In 2014-2015, Egypt faced rising poverty, and high inequality between regions, with over 68% of energy subsidies benefiting the richest two quintiles.

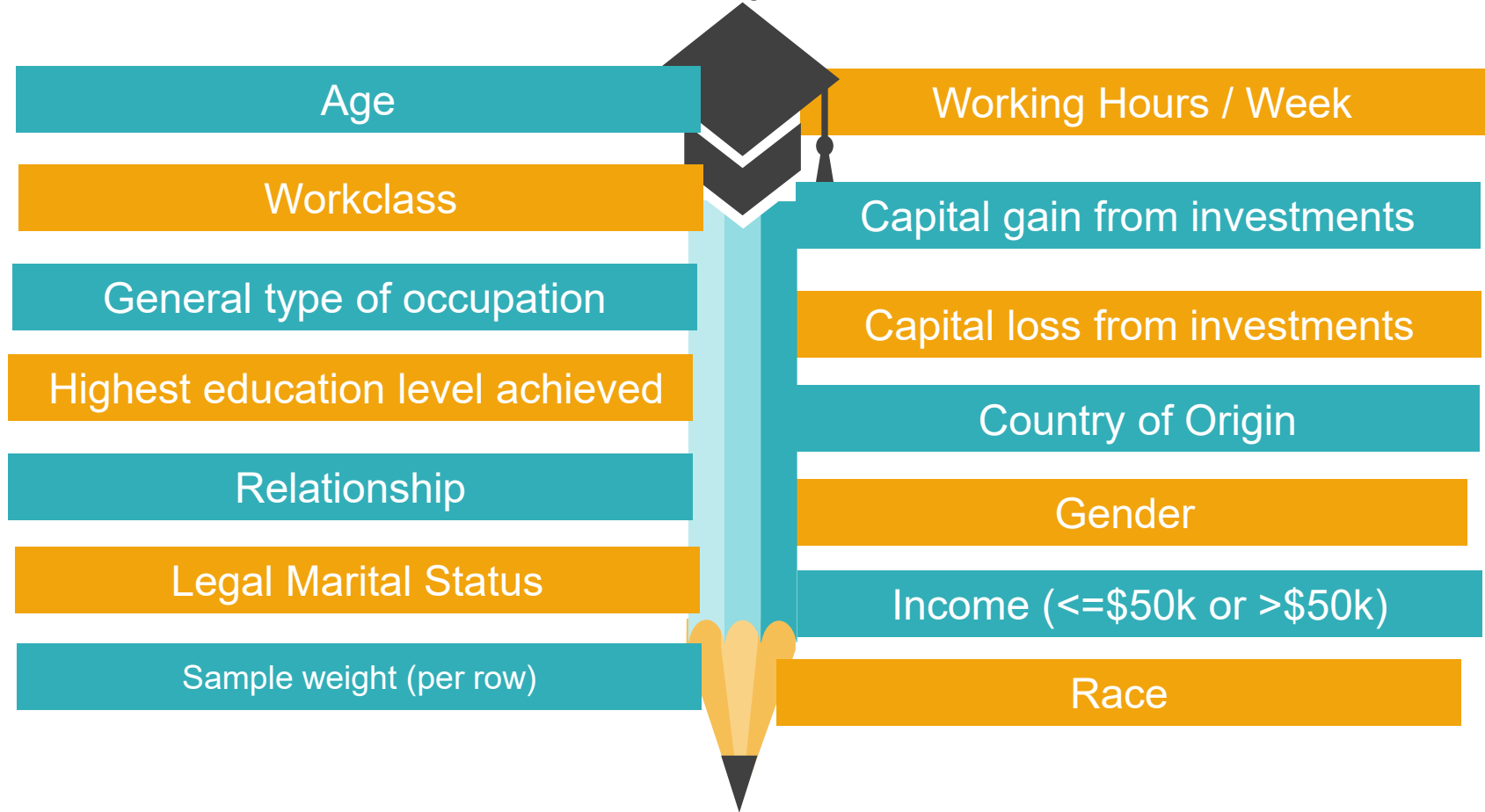
The government moved to **targeted cash transfers**, to determine who qualifies as poor and non-poor.

This system used household data—such as family size, education, assets, housing quality, and demographics—to predict income category (eligible vs. ineligible for aid).



Source: World Bank. (2015). *Project Appraisal Document: Strengthening Social Safety Net Project / Takaful & Karama*

Data Summary & Overview



Data Cleaning

01

Standardized

Standardized column names and made categories more interpretable

02

Merged Data

Grouped related categories in key features

03

Ignoring Unnecessary Columns

Disregarded unnecessary columns

Feature Engineering

01

Flagged for Investments

Created a column that is flagged 1 if the sample has capital activity

02

Computed Data

Calculated a combined “net_capital” feature from gains and losses

03

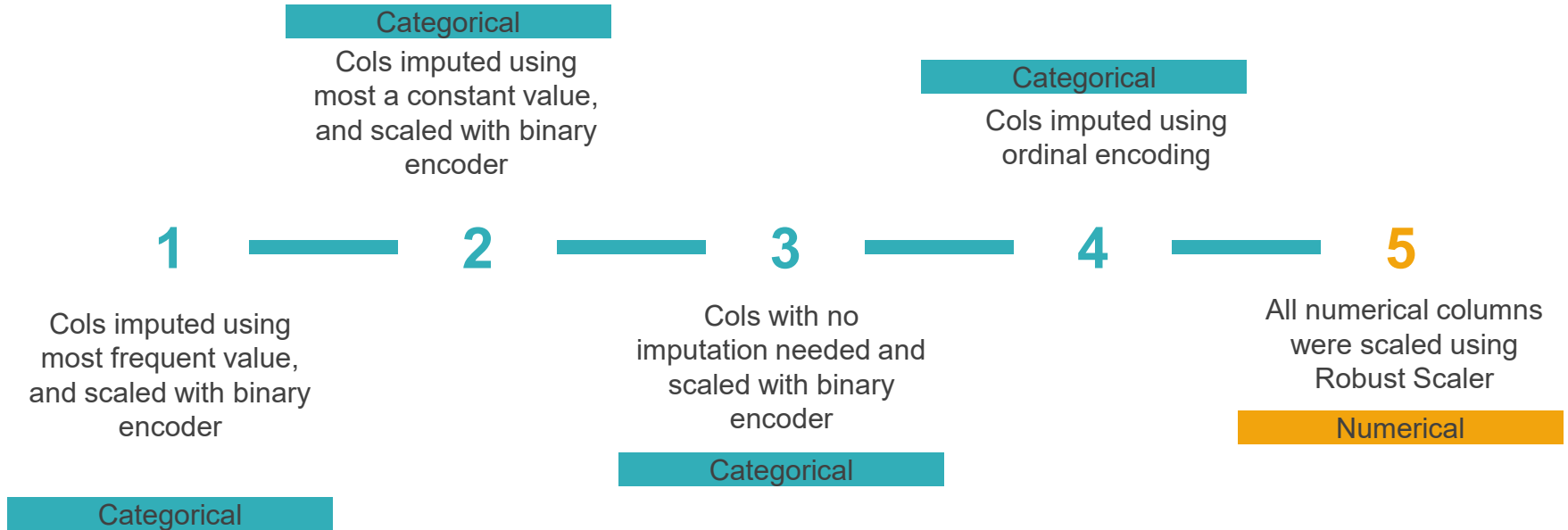
Grouped Data

Computed a column with the employment type (full-time/part-time/etc) based on the working hours per week



Data Visualization

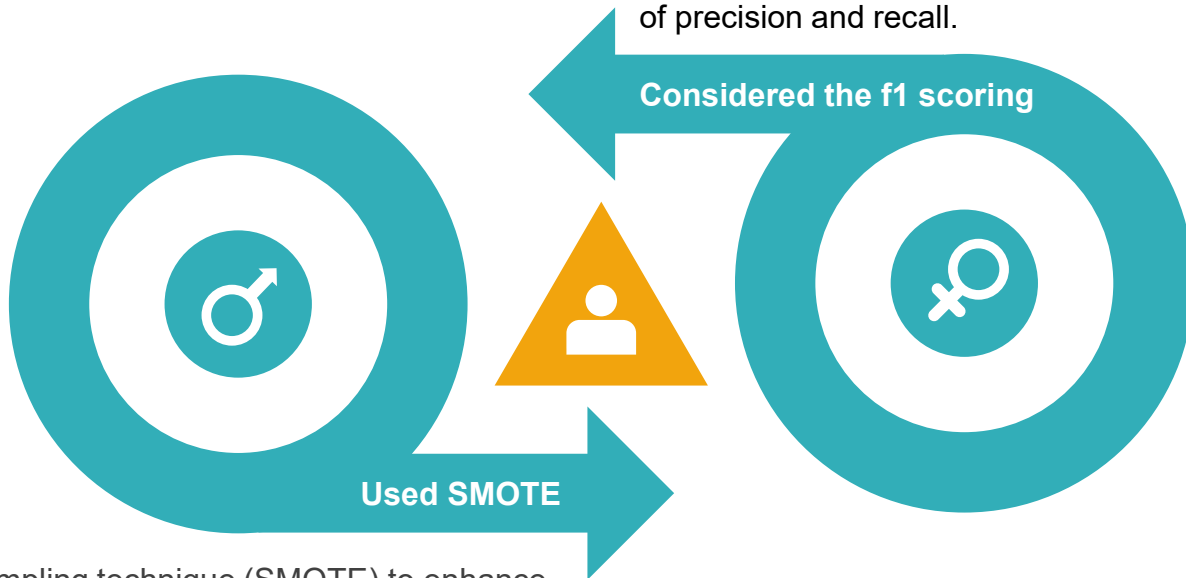
Data Preprocessing Pipeline



Handling Imbalance

Almost 76% of the data were getting paid less than or equal \$50k compared to 23.9% getting paid over \$50k a year

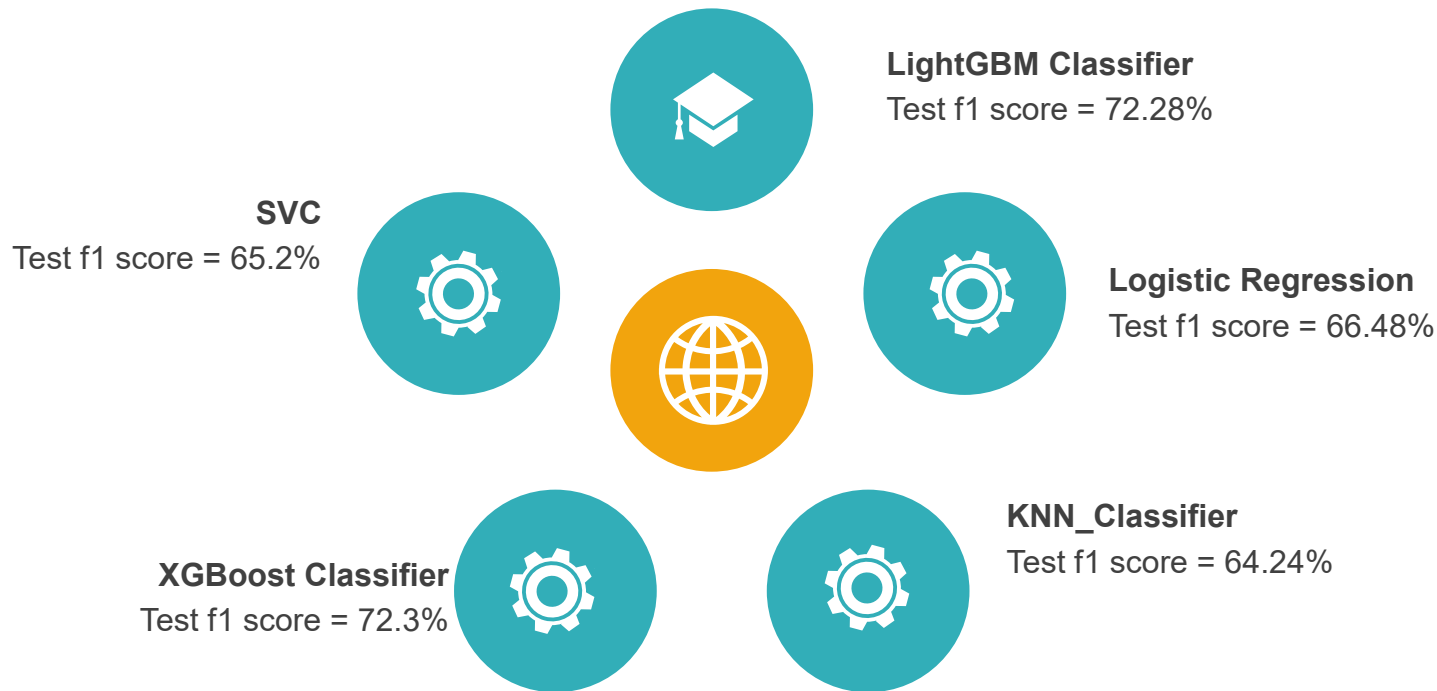
The F1 score provides a more accurate measure of the model performance, as it represents the harmonic mean of precision and recall.



Used the oversampling technique (SMOTE) to enhance the model prediction of the smaller class

Modeling

Trials: Best Fitting Model



Hyperparameter Tuning: LightGBM Classifier

Used the RandomizedSearchCV technique.

Round #1

Test f1 score

70.61%



Round #2

Test f1 score

71.92%



Round #3

Test f1 score

72.17%



Result

Hyperparameter tuning did not improve the model, as the initial F1 score of 72.28% remained the best, while tuned models performed worse.

Feature Selection

Wrapper Method

KBestSelector

Identified the most predictive features using ANOVA F-scores.

Features Selected

Top 25

Chose the top 25 highest-scoring features in an attempt to maximize model relevance.

Model Performance

69.25%

Feature selection resulted in a lower F1 score



Conclusion

Neither the hyperparameter tuning nor the wrapper feature selection methods resulted in an enhanced model, and the default model turned out to be the best



Deployment



Thank You!